

[DC] Foveated Fluid Animation in Virtual Reality

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ABSTRACT

Large-scale fluid simulation is often utilized in virtual reality (VR) applications. Physics-based fluid simulation benefits computer graphics with more realistic fluid phenomena, increasing users' immersion in VR. However, the requirements of real-time implementation and the limited computation resources hinder its application in VR. In this article, we first propose foveated fluid animation for applying physics-based fluid simulation in VR. We divide the entire field of view (FOV) into three eccentric circular regions, and fluid particles are scaled into three sizes adaptively in terms of their distributions from the foveal to the peripheral. The proposed method is expected to provide VR applications with large-scale fluid simulation while maintaining high simulation accuracy and minimal computation load. Furthermore, we introduce a novel metric, Fluid Particle Per Degree (FPPD), for optimizing the performance of the proposed method. Both subjective and objective measurements are expected to conduct based on the prototype in the future. We expect the metric to benefit the application of foveated fluid animation with optimal performance in future VR.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality; Computing methodologies—Computer graphics—Animation—Physical simulation;

1 INTRODUCTION

Large-scale fluid simulation is essential for a variety of virtual reality (VR) applications such as disaster education [7], sports training [3], entertainment [6]. However, existing fluid simulation in VR appears less realistic due to the lack of surface details of the fluid, reducing the users' immersion significantly. Physics-based animation generates realistic fluid details without extra artistic editing. A few works have been proposed to develop VR applications based on physics-based fluid simulation. Chen et al. [1] simulate the concentrated sulfuric acid dilution experiment using particle-based fluid simulation. Nevertheless, all the previous works target small-scale fluid. Implementing physics-based animation on large-scale fluid simulation in VR is challenging due to the heavy computation load.

The adaptive method allocates computing resources to regions of the fluid with different scales adaptively to mitigate the computation load of physics-based fluid simulation efficiently [8]. Regarding the non-uniform visual acuity of human beings, the foveated rendering technique is similarly utilized to accelerate the frame rendering in VR by adaptively omitting the unperceived detail in the peripheral vision [2].

Motivated by the adaptive fluid simulation method and the foveated rendering technique, we first propose foveated fluid animation for large-scale fluid simulation in VR in this article. We focus on the Lagrangian particle-based method, which has been widely used in real-time fluid applications [5]. Fluid particles in the virtual environment are adaptively sampled while keeping sufficient

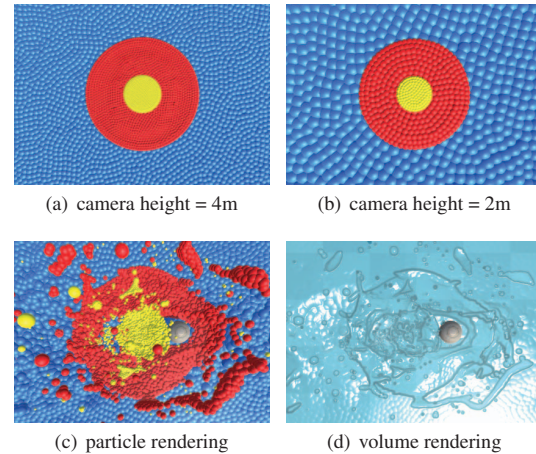


Figure 1: Pseudo-foveated fluid animation results.

simulation accuracy. Thus, the problem caused by the exponential growth of particle numbers in higher resolution is expected to be solved. We divide the visible domain into three regions in the foveated fluid animation. The innermost region is simulated at the highest resolution, while the outer peripheral regions are simulated with progressively lower resolution.

Furthermore, a novel metric, fluid particles per degree (FPPD), is proposed to analyze the distribution of fluid particle radius spanning the user's field of view (FOV). We expect the quantified function between FOVs and proper FPPD is developed based on future user studies, which may benefit the proposed method with an optimal trade-off between the workload and the animation quality.

2 PROPOSED METHOD

2.1 Foveated Fluid Simulation

The proposed simulation method is based on the Position Based Fluids (PBF) solver [4]. Similar to the user's view is divided into three eccentricity layers with steeply decreased pixel density from his/her gaze point in the foveated rendering technique, we also divide the visible domain into three different regions in the foveated fluid animation, as shown in Figure 1(a). Simulation and rendering for fluid particles in each region are conducted simultaneously but with different fluid particle scales.

We denote the three simulation targets as inner, middle, and outer regions. The inner region corresponding to the foveal region of human vision requires the finest simulation details. Thus, we use the smallest particle in the inner region to run the high-resolution simulation (yellow particles). The middle region is also a square area centered about the gaze point, whose radius is set to three times of inner region's. Furthermore, the outer region covers the rest of the area. Then, simulations of the middle and the outer regions are conducted with a median (red particles) and the largest (blue particles) fluid particle sizes, respectively, leading to the less realistic fluid phenomenon with a softer computation load.

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Moreover, the three-scale particle simulation demands a coupling algorithm to modify and smooth the inconsistent flow behaviors that might happen at the boundary of two different regions. The inner area needs the outer area to provide the appropriate boundary condition, while the outer area needs the inner one to provide feedback information to maintain the consistency of fluid motions.

2.2 Preliminary Experiment

As shown in Figure 1, we conducted a pseudo-foveated fluid animation in the preliminary experiment. We place the camera above the fluid particles and fix the gaze point at the center of the screen. To ensure the consistency of different scale fluids' behaviors, we let the three simulations keep the same parameter except for the particle radii. Besides, we use the screen-space fluid rendering method for rendering.

With the camera's height decreasing from 4m to 2m, fluid particles appear more prominent while the angular sizes of the three visual regions remain constant, as shown in Figure 1(a) and Figure 1(b). However, the foveated method is expected to adjust the particle radii automatically to remain the same particle size on the screen. The dynamics of a stone hitting the water tank are presented in Figure 1(c) and Figure 1(d) with different rendering styles. Due to the downsized fluid particles, more vivid splashes are generated in the inner region than in the middle region when the stone hits the boundary of regions.

3 METRIC FOR FLUID SIMULATION IN VR

It is common to adjust the particle radius to balance the simulation accuracy and computation load in typical physics-based fluid simulation. Exploring appropriate fluid particle radii for different regions is crucial for the proposed method to optimize performance. However, the metric of the particle radius is no longer practical in VR applications. A small particle radius is required at the inner region to guarantee fine details for a user close to the fluid, but the same radius is not feasible when the user leaves the fluid away since the fluid particle number becomes overwhelmingly high, which causes a hefty computation load. Therefore, we propose the novel metric, FPPD, to assist the balance between the simulation accuracy in a user's vision and the system workload in VR applications.

Similar to Pixels Per Degree (PPD) for pixel density, FPPD is calculated by:

$$FPPD = \frac{N_{particle}}{\theta} = \frac{w}{2r\theta} \quad (1)$$

where r is the fluid particle radius, w is the width of the fluid region. Moreover, θ represents the FOV, which can be computed via:

$$\theta = 2\arctan\left(\frac{w}{2d_{view}}\right) \quad (2)$$

where d_{view} is the viewing distance.

The crown splash scenes in Figure 2 show three single-scale fluid simulations with corresponding FPPD values of three regions in Figure 1(a), which are calculated as 1.4035, 0.7297, and 0.4444, respectively. The simulation with the smallest FPPD cannot achieve a reasonable splash result, as shown in Figure 2(b). Although the medium FPPD value can produce a splash, its water surface still appears slightly bumpy, as shown in Figure 2(c). In Figure 2(d), the simulation with the inner region FPPD value can produce the best interface details and a smooth surface.

Moreover, we speculate that the difference between fluid particles and pixels causes the numerical gap. For example, particle overlaps occur during the simulation, increasing the FPPD value. Besides, the dynamic particle and static particles also perform differently.

In summary, the function between various FOVs and the corresponding FPPD values should be built for optimizing system performance. Both subjective and objective measurements will be con-

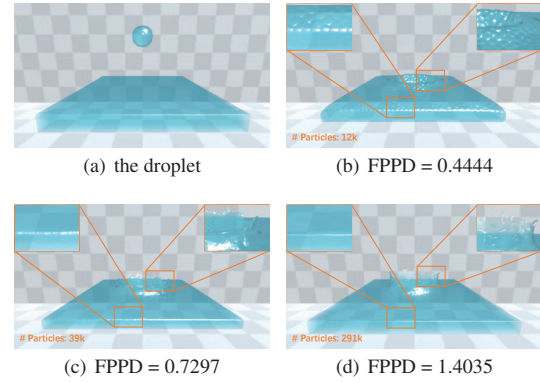


Figure 2: Crown splash with corresponding FPPD values of three regions in Figure 1(a)

ducted to develop the proposed metric as guidance for the practical application of the foveated fluid animation method.

4 EXPECTED CONTRIBUTIONS AND FUTURE WORK

Foveated Simulation: By exploiting foveation in fluid simulation, the animation result would look like a full-resolution animation but reduce the number of fluid particles immensely. However, the saccades might cause unexpected artifacts after adding the eye-tracking technique to our method. Besides, unlike blending images in foveated rendering, combining three different scale simulations requires a more sophisticated coupling algorithm.

Fluid Particles Per Degree: We propose the metric for fluid simulation in VR to assist in balancing the accuracy and workload. The problem of moving the camera we mentioned in section 2.2 is solvable by adjusting the fluid scales to keep the same FPPD in each region all the time. As the pixel in the screen is not equivalent to the particle in the fluid simulation, it is essential to investigate more factors that might affect FPPD.

The next step of our research will focus on implementing foveated gaze-contingent simulation method and designing some user studies to validate our results and refine the formula of FPPD. Another interesting direction for future work is to interact with fluids in VR. Furthermore, a more efficient real-time fluid rendering method would improve our animation.

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